

Carbohydrates Metabolism: Glycolysis

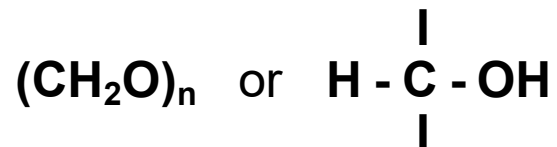
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Objectives:

- At the end of this chapter, it is expected to;
 - State the Biomedical Importance of Glycolytic pathway
 - Describe the chemical reactions of Glycolysis
 - Understand the regulations of the Glycolytic pathway
 - Explain the Importance of glycolysis for normal red blood cells
 - Explain the consequences of deficiencies in glycolytic enzymes
 - Calculate the energetics of glycolytic pathway
 - Explain the Warburg effect in Cancer cells

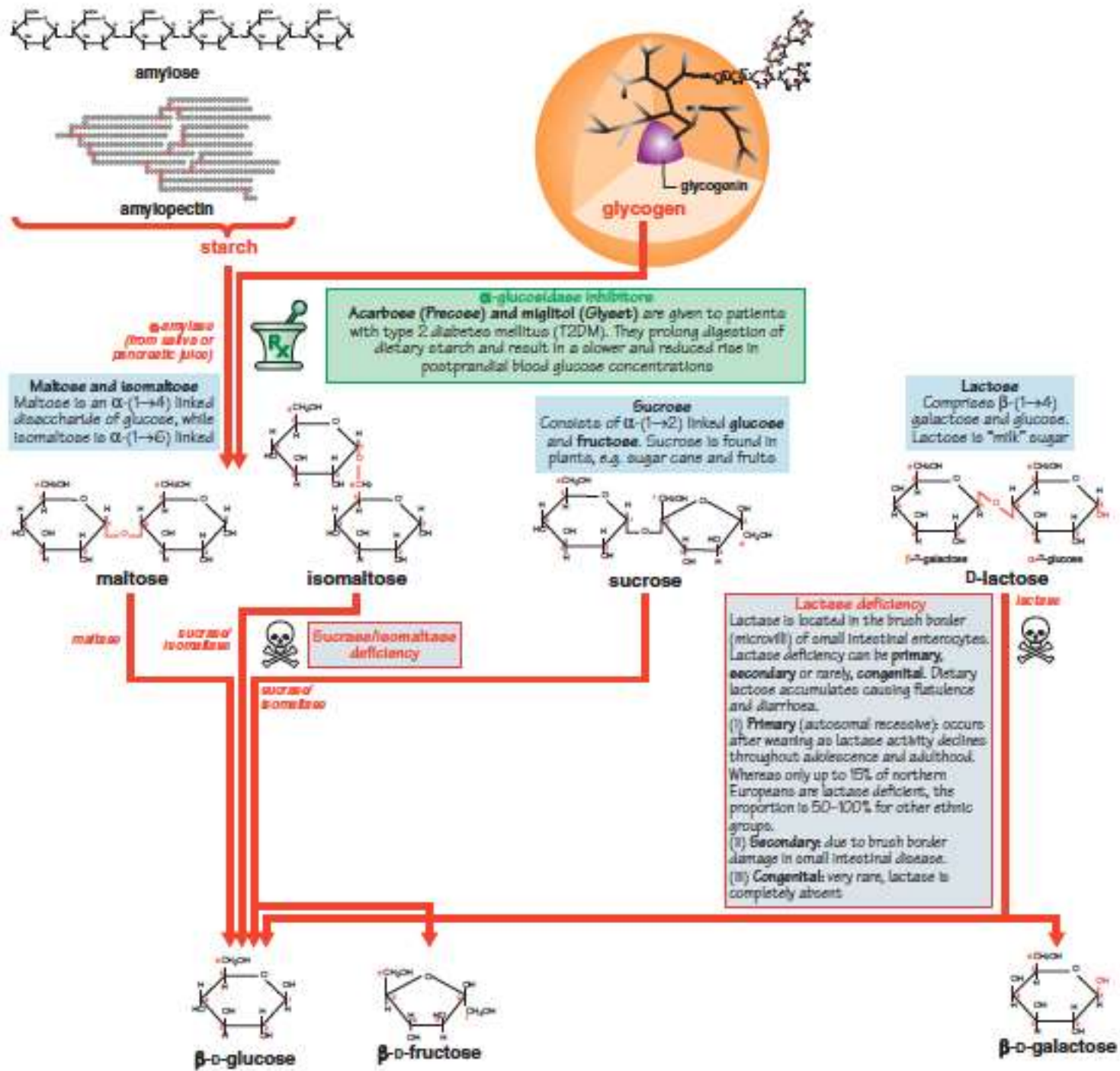
Carbohydrates

- Major source of energy from our diet
- Composed of the elements C, H and O



- Produced by photosynthesis in plants
- It (glycans) includes the following:
 - ◆ Monosaccharides
 - ◆ Disaccharides
 - ◆ Oligosaccharides
 - ◆ Polysaccharides

Digestion and Absorption of carbohydrates

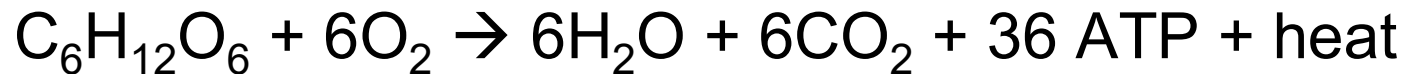


Family of glucose transporters

Name	Tissue location	K_M	Comments
GLUT1	All mammalian tissues	1 mM	Basal glucose uptake
GLUT2	Liver and pancreatic β cells	15–20 mM	In the pancreas, plays a role in the regulation of insulin In the liver, removes excess glucose from the blood
GLUT3	All mammalian tissues	1 mM	Basal glucose uptake
GLUT4	Muscle and fat cells	5 mM	Amount in muscle plasma membrane increases with endurance training
GLUT5	Small intestine	—	Primarily a fructose transporter

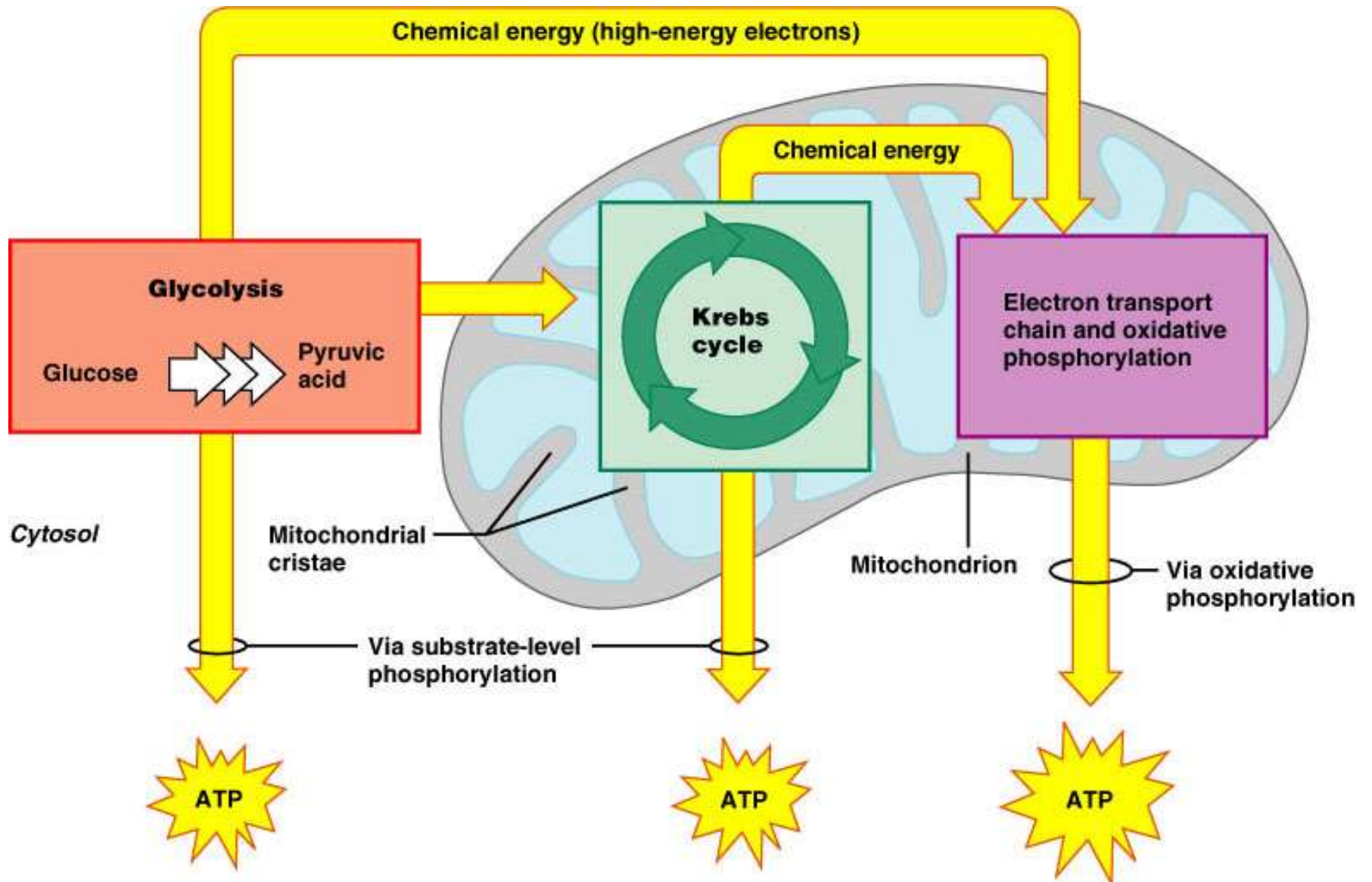
Carbohydrate Metabolism

- Since all carbohydrates are transformed into glucose, it is essentially glucose metabolism
- Oxidation of glucose is shown by the overall reaction:

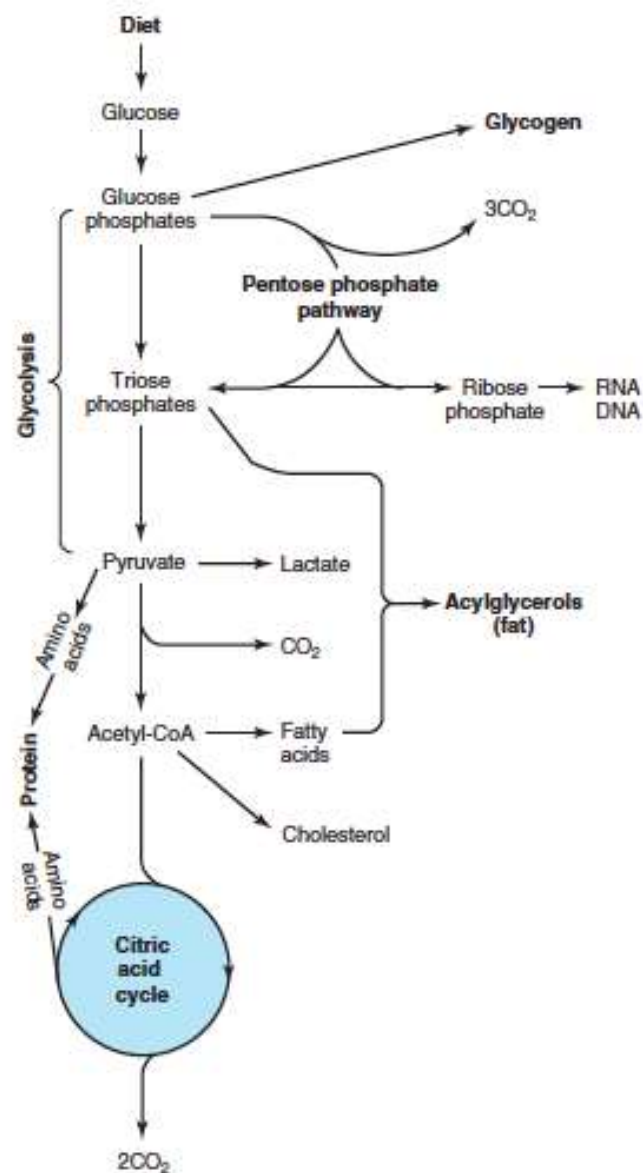


- Glucose is catabolized in three pathways
 - Glycolysis
 - Krebs cycle
 - The electron transport chain and oxidative phosphorylation

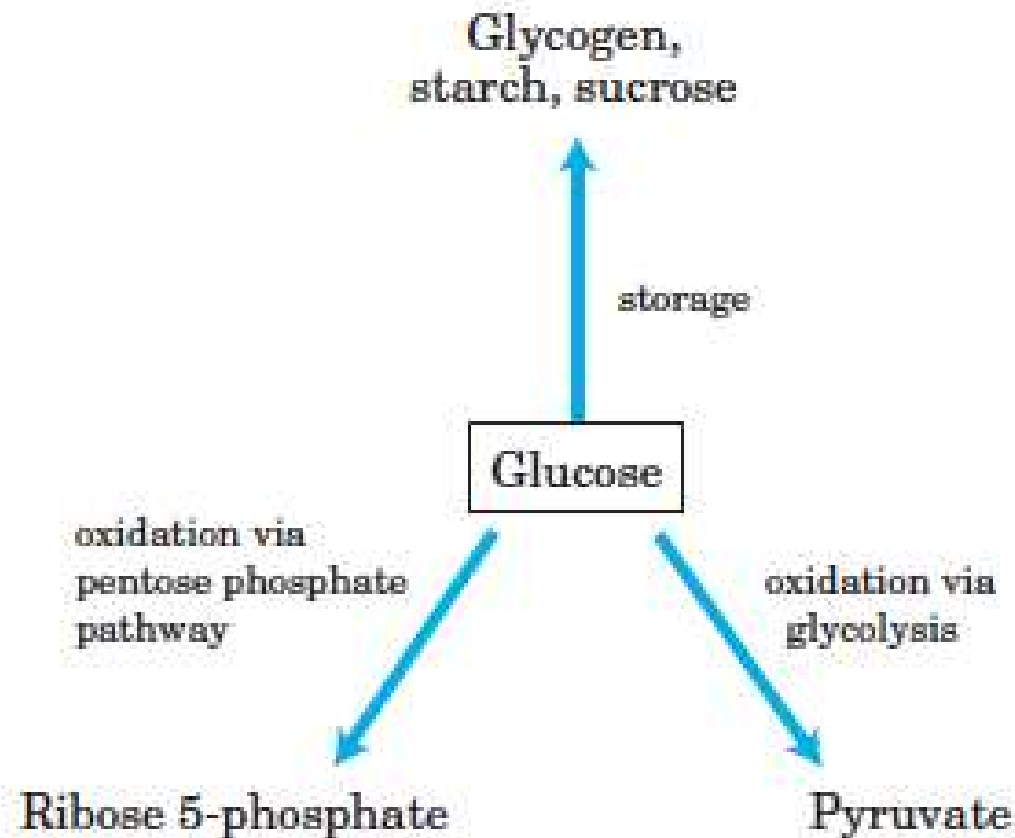
Carbohydrate Catabolism



Overview of carbohydrate metabolism



Major pathways of glucose utilization



GLYCOLYSIS:

- The major pathway for glucose metabolism

Biomedical Importance Of Glycolysis

- **Major pathway for glucose metabolism,**
- Occurs in the cytosol of all cells.
- Main pathway for the metabolism of **fructose, galactose, and other dietary carbohydrates** also.
- It is unique - can **function either aerobically or anaerobically**
- This pathway is meant for **provision of energy.**
- **Erythrocytes**, which lack mitochondria, are completely reliant on glucose as their metabolic fuel, and metabolize it by anaerobic glycolysis
 - **Haemolytic anaemias:** Inherited enzyme deficiencies like hexokinase deficiency and pyruvate kinase deficiency can produce haemolytic anaemia.

Biomedical Importance Of Glycolysis (Cont.)

- In skeletal muscle, **provides ATP even in absence of O₂**.
 - Muscles can survive anoxic episodes.
- **Heart muscle:** As compared to skeletal muscle, heart muscle is adapted for aerobic performance.
 - It has relatively poor glycolytic activity and poor survival under conditions of ischaemia.
- **Role in cancer therapy:** In fast-growing cancer cells, rate of glycolysis is very high.
 - Produces more pyruvic acid (PA) than TCA cycle can handle.
 - Accumulation of pyruvic acid leads to excessive formation of lactic acid producing local lactic acidosis.
 - Local acid environment may be congenital for certain cancer therapy.

GLYCOLYSIS

- Also called as **Embden Meyerhof pathway**.
- Glykys (Greek) meaning “sweet” and lysis meaning “splitting”.
- Series of enzyme-catalyzed reactions
- The breakdown of the 6 ‘C’ glucose into two molecules of the 3 ‘C’ pyruvate occurs in ten steps
 - The first 5 of which constitute the preparatory phase and the rest 5 constitute the payoff phase of glycolysis

Overall reaction of glycolysis:



Glycolysis

- A **three-phase pathway** in which Glucose is oxidized into pyruvic acid (PA)
 - Phase 1: **Sugar activation**
 - Phase 2: **Sugar cleavage** (splitting)
 - Phase 3: **Oxidation and ATP formation**
- Pyruvic acid: end-product of glycolysis
 - In aerobic condition, moves on to the Krebs cycle
 - In anaerobic environment, is reduced to lactic acid

pyruvic acid $\longleftrightarrow$ lactic acid

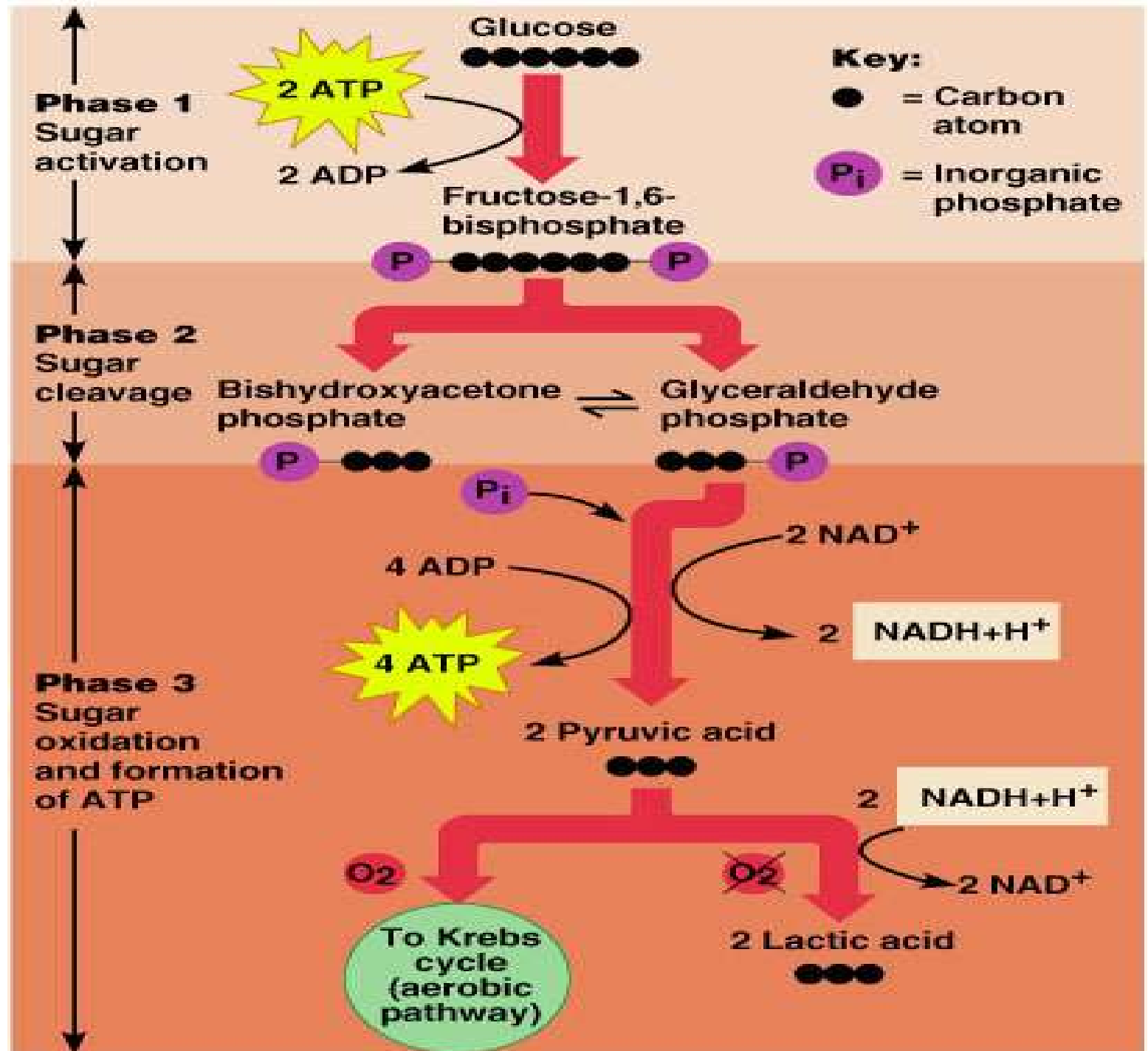
Glycolysis: Phase 1 and 2

- **Phase 1: Sugar activation**
 - Two ATP molecules activate glucose into fructose-1,6-diphosphate
- **Phase 2: Sugar cleavage (splitting)**
 - Fructose-1,6-bisphosphate (6 C' s) is split into two 3 'C' compounds, i.e., Glyceraldehyde 3-phosphate and DHAP

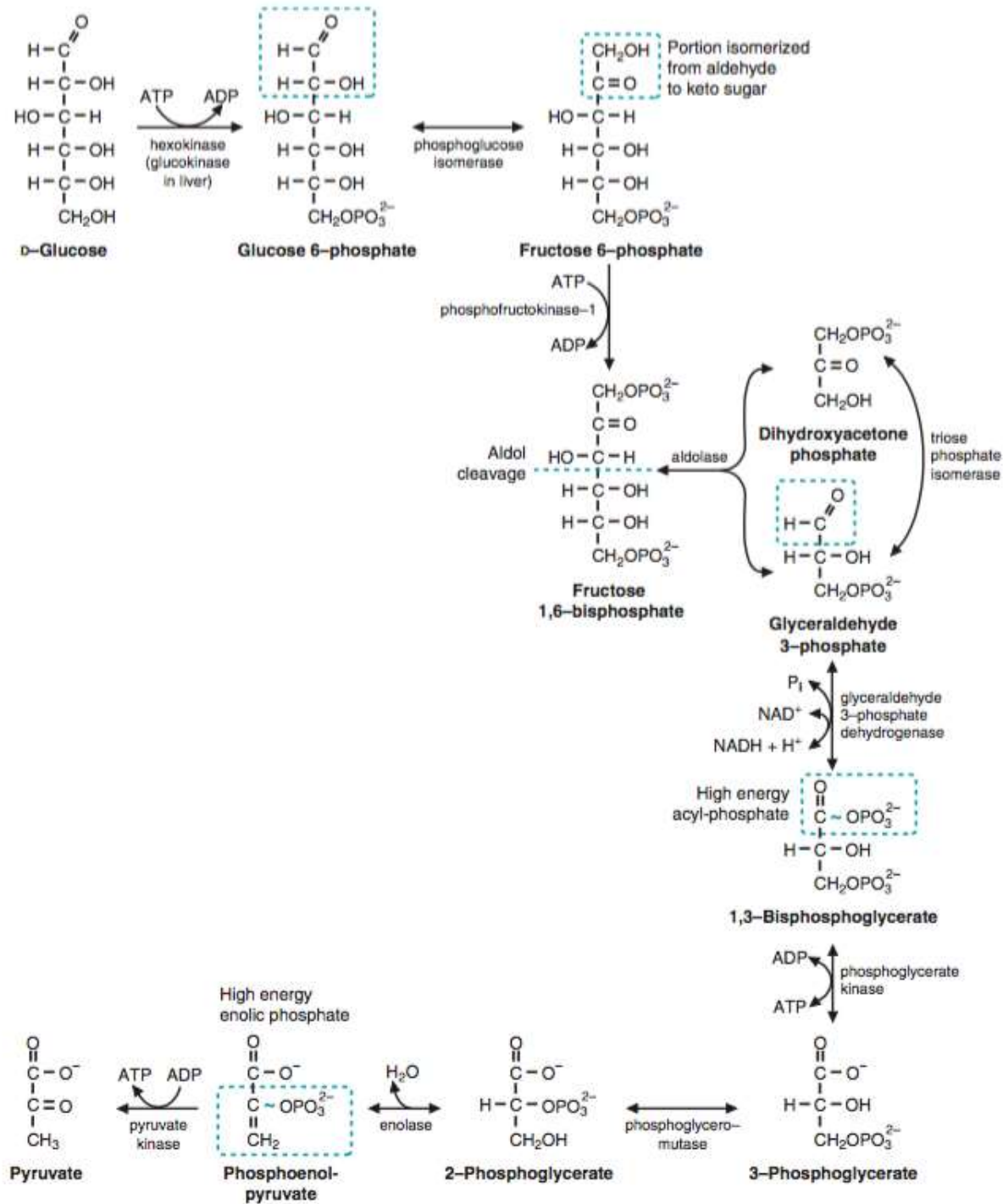
Glycolysis: Phase 3

- **Phase 3: Oxidation and ATP formation**
 - The 3-carbon sugars are oxidized (reducing NAD^+); i.e., $2 \text{H}' \text{ s} + \text{NAD} \longrightarrow \text{NADH}_2$
 - Inorganic phosphate groups (P_i) are attached to each oxidized fragment
 - The terminal phosphates are cleaved and captured by ADP to form four ATP molecules
 - The final products are:
 - Two pyruvic acid molecules
 - Two $\text{NADH} + \text{H}^+$ molecules (reduced NAD^+)
 - A net gain of two ATP molecules

Glycolysis

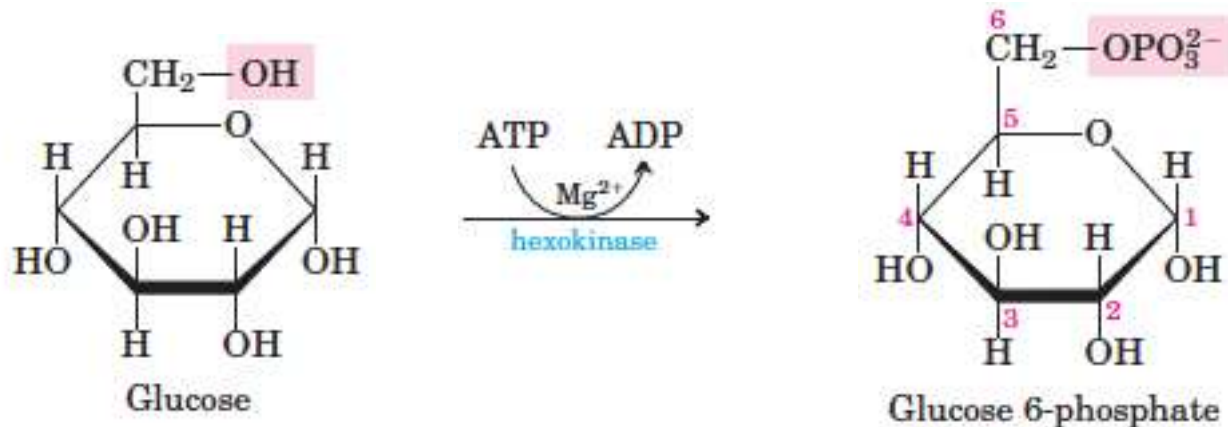


Reactions of Glycolysis



STEPS OF GLYCOLYSIS

1. Phosphorylation of Glucose



Hexokinase is present in all cells of all organisms.

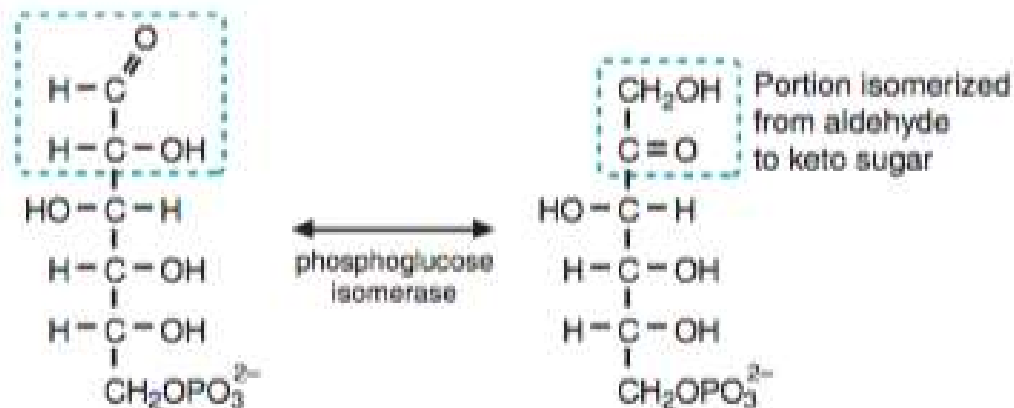
Hepatocytes also contain a form of hexokinase called hexokinase IV or glucokinase.

Both are isozymes.

Differences between hexokinase and glucokinase

Hexokinase	Glucokinase
1. Non-specific, can phosphorylate any of the hexoses 2. More stable 3. Found almost in all tissues 4. Found in foetal as well as in adult liver 5. Allosteric inhibition by glucose-6-P 6. K_m is low = 0.1 mM, hence high affinity for glucose 7. Not very much influenced by diabetic state/or fasting 8. No change with glucose feeding 9. Inhibited by glucocorticoids and GH; insulin does not have effect on hexokinase 10. Hexokinase activity of liver found in three enzyme proteins (isoenzymes) 11. <i>Main function to make available glucose to tissues for oxidation at lower blood glucose level</i>	1. Specific, can phosphorylate glucose only 2. Physiologically more labile 3. Found only in liver 4. Found in adult liver, not in foetal liver 5. Not inhibited by Glucose-6-P 6. K_m is high = 10 mM, low affinity for glucose 7. Depressed in fasting and in diabetes. Glucokinase is deficient in patients of DM, changes according to nutritional status 8. Increased by feeding of glucose after fasting 9. Inhibited by glucocorticoids and GH; glucose and insulin stimulates. Synthesis is induced by insulin, an <i>inducible enzyme</i> 10. Not known 11. <i>Main function to clear glucose from blood after meals and at blood levels greater than 100 mg/dl</i>

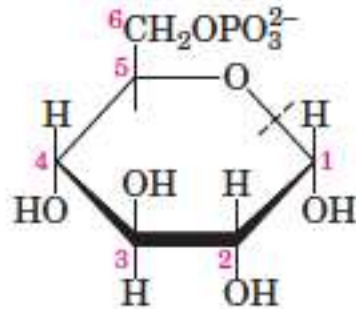
2. Conversion of Glucose 6-Phosphate to Fructose 6-Phosphate



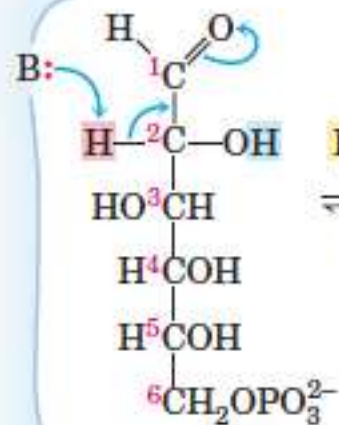
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The phosphohexose isomerase reaction

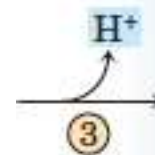
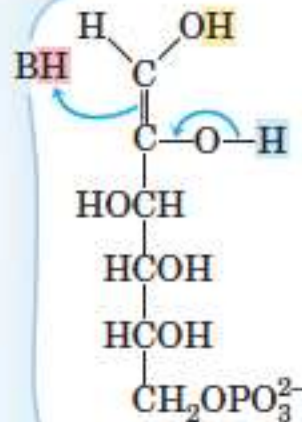
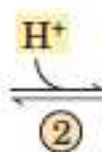
Glucose 6-phosphate



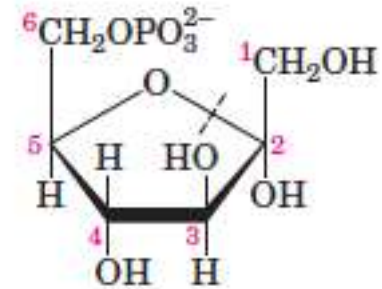
① binding and ring opening



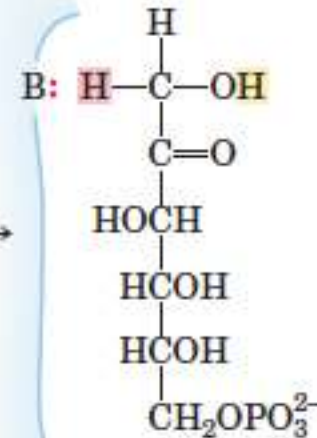
Phosphohexose isomerase



Fructose 6-phosphate



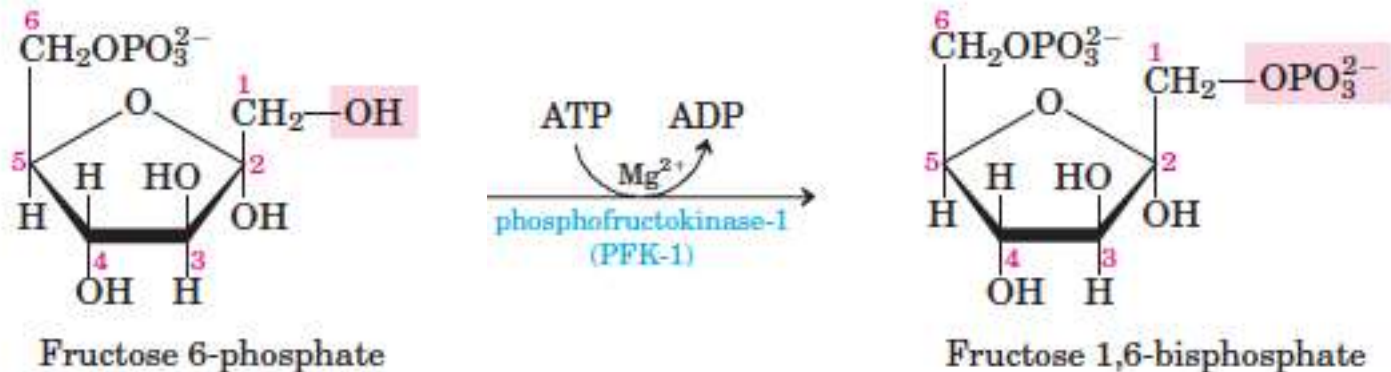
④ ring closing and dissociation



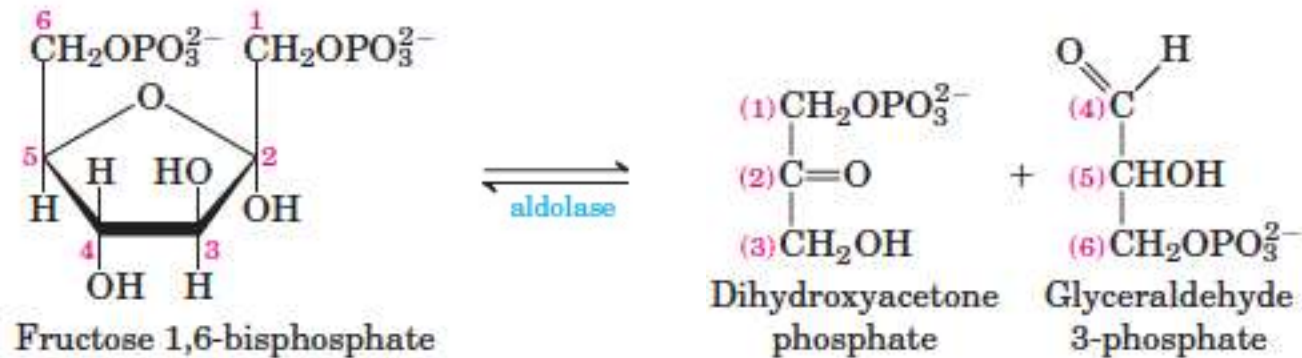
The phosphohexose isomerase reaction

- The ring opening and closing reactions (steps 1 and 4) are catalyzed by an active-site His residue.
- The movement of the proton between C-2 and C-1 (steps 2 and 3) is base-catalyzed by an active-site Glu residue (shown as B:).
 - The proton (pink) initially at C-2 is made more easily abstractable by electron withdrawal by the adjacent carbonyl and the nearby hydroxyl group.
 - After its transfer from C-2 to the active-site Glu residue, the proton is freely exchanged with the surrounding solution; that is, the proton abstracted from C-2 in step 2 is not necessarily the same one that is added to C-1 in step 3.
 - The additional exchange of protons (yellow and blue) between the hydroxyl groups and solvent is shown for completeness. The hydroxyl groups are weak acids and can exchange protons with the surrounding water whether the isomerization reaction is underway or not.

3. Phosphorylation of Fructose 6-Phosphate to Fructose 1,6-Bisphosphate

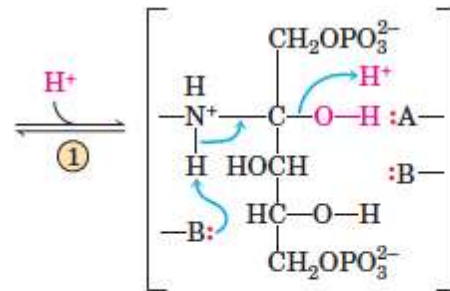
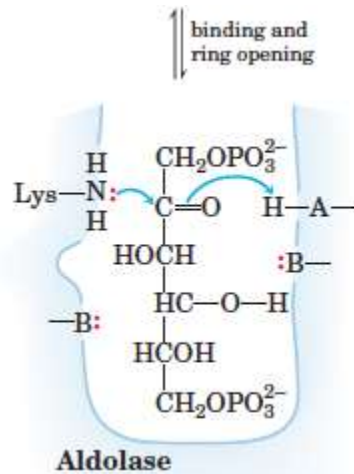
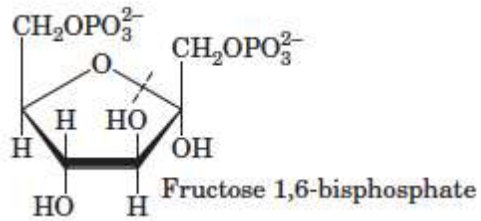


4. Cleavage of Fructose 1,6-Bisphosphate

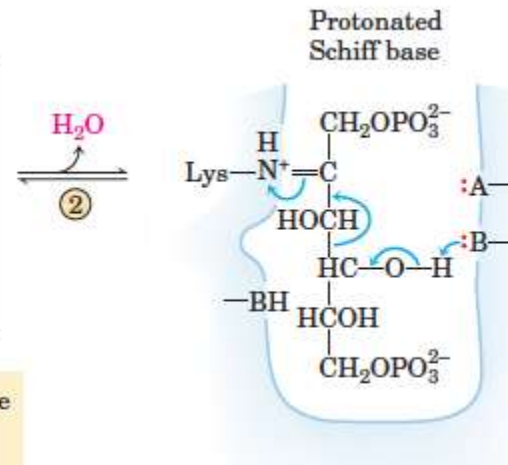


Class I aldolases, found in animals and plants
Class II enzymes, in fungi and bacteria

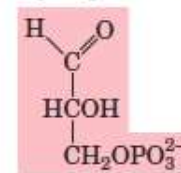
The class I aldolase reaction



In steps ① and ② an amine is converted to a Schiff base (imine).

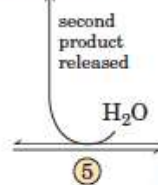
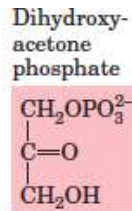
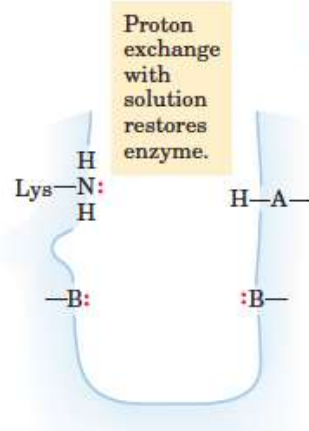


Glyceraldehyde 3-phosphate

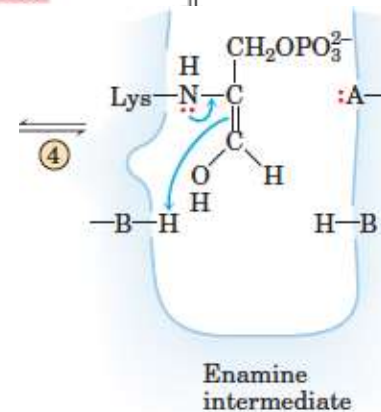
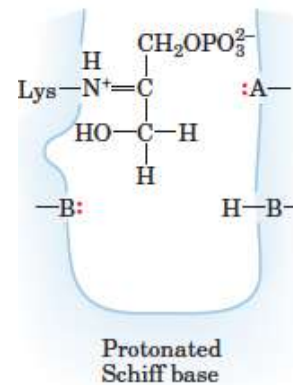


first product released

③



Schiff base is hydrolyzed in reverse of Schiff base formation.

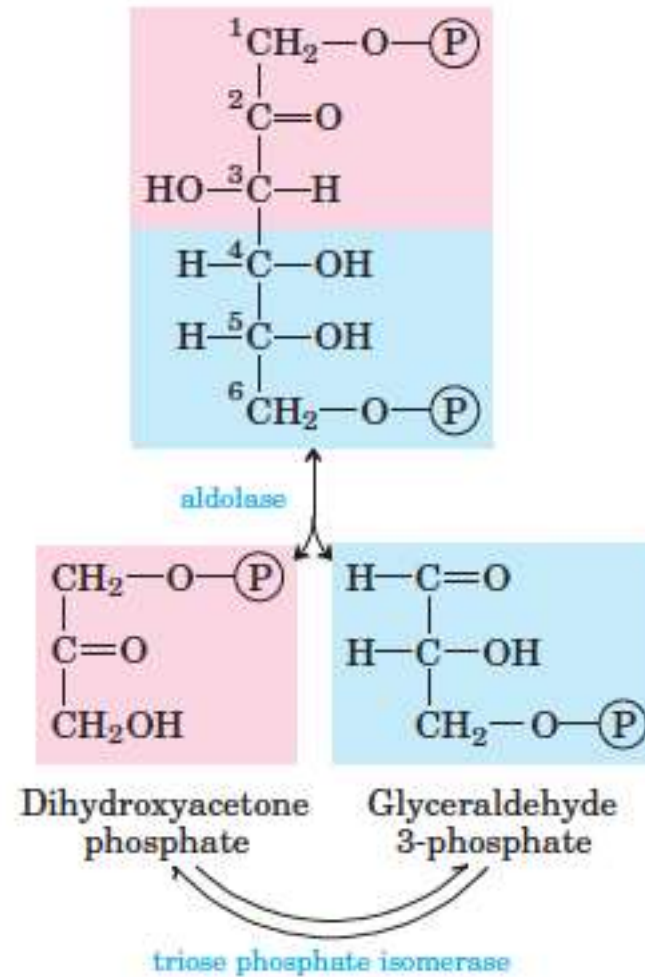


The class I aldolase reaction

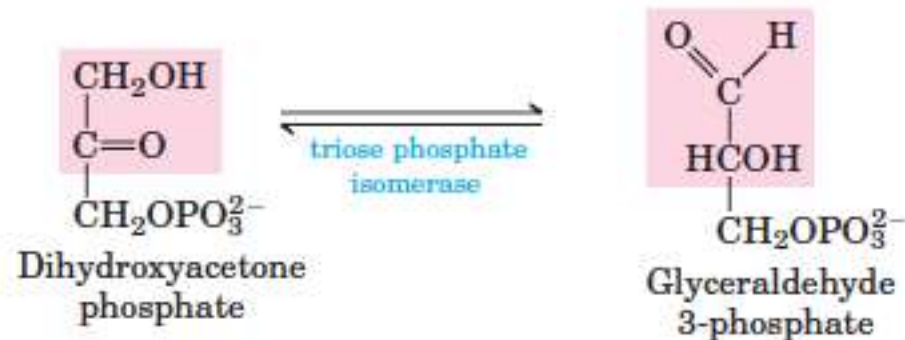
- The reaction shown here is the reverse of an aldol condensation.
- The cleavage between C-3 and C-4 depends on the presence of the carbonyl group at C-2.
- Step 1 and 2: The carbonyl reacts with an active-site Lys residue to form an imine, which stabilizes the carbanion generated by the bond cleavage—an imine delocalizes electrons even better than does a carbonyl.
- Step 3: Bond cleavage releases glyceraldehyde 3-phosphate as the first product.
- Step 4: The resulting enamine covalently linked to the enzyme is isomerized to a protonated Schiff base, and (Step 5) hydrolysis of the Schiff base generates dihydroxyacetone phosphate as the second product.
- A and B represent amino acid residues that serve as general acid (A) or base (B).

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Fructose 1,6-bisphosphate



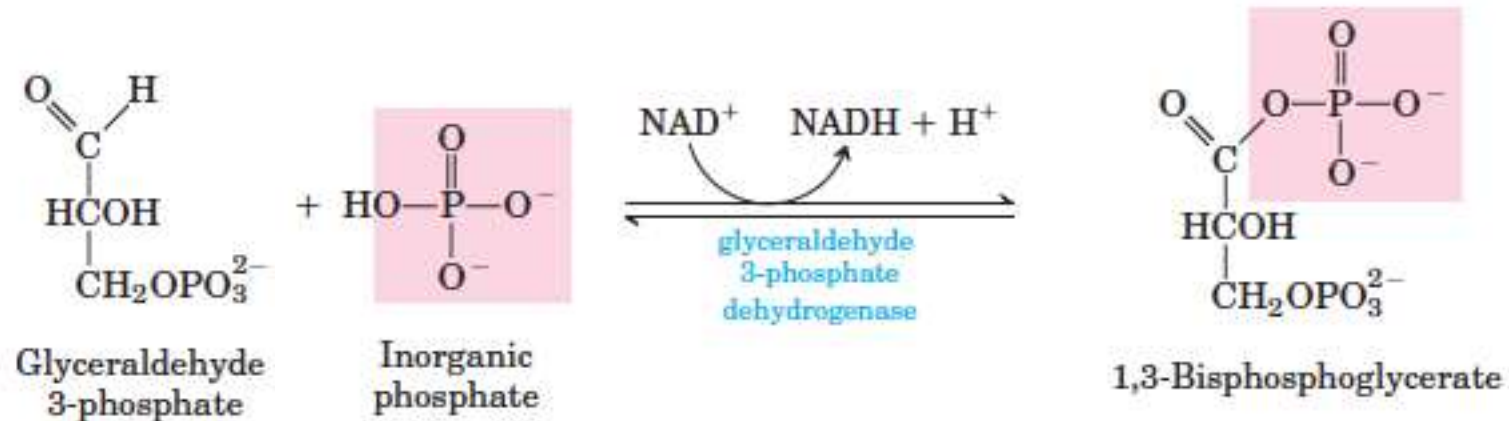
5. Interconversion of the Triose Phosphates



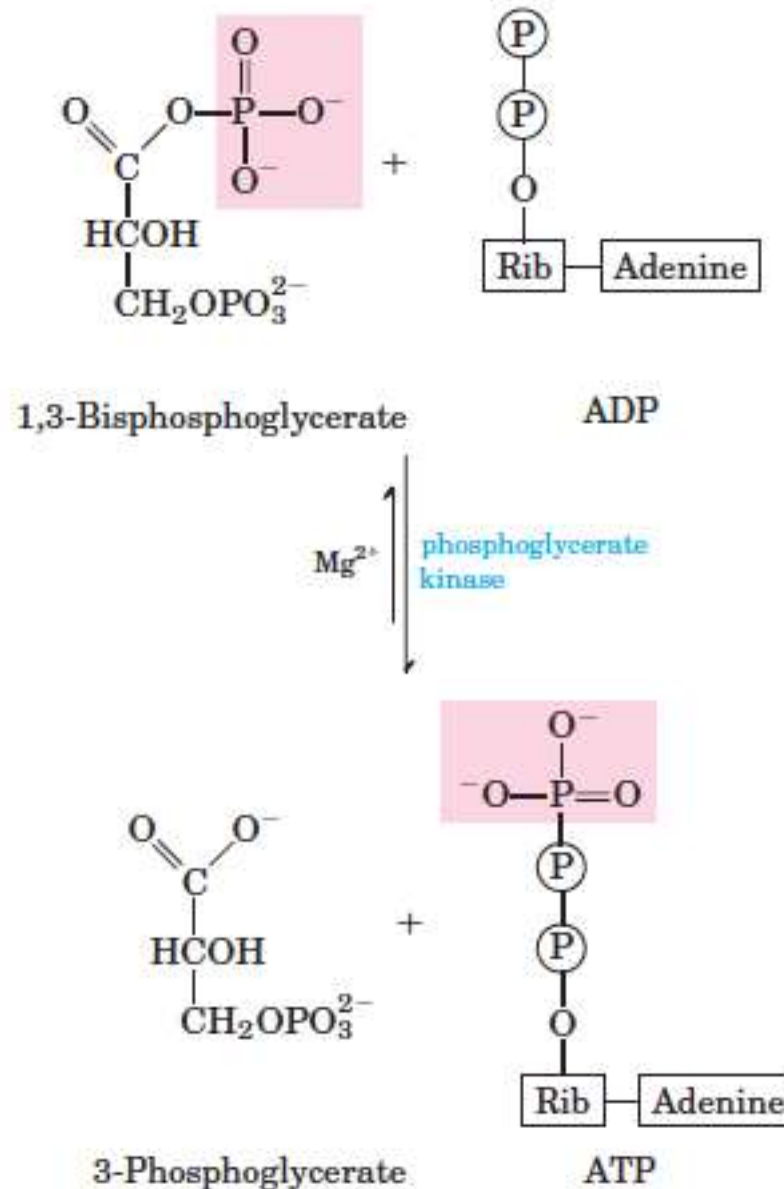
CLINICAL CORRELATES

Patients with **triose phosphate isomerase (TPI)** deficiency have neonatal-onset **hemolytic anemia** as well as progressive **neurologic involvement**. Children have progressive hypotonia with eventual diaphragm paralysis that requires ventilation, as well as **cardiomyopathy**.

6. Oxidation of Glyceraldehyde 3-Phosphate to 1,3-Bisphosphoglycerate

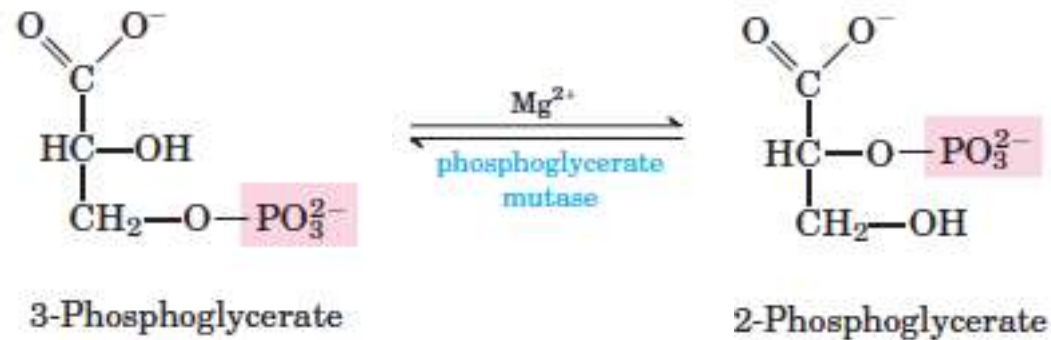


7. Phosphoryl Transfer from 1,3-Bisphosphoglycerate to ADP

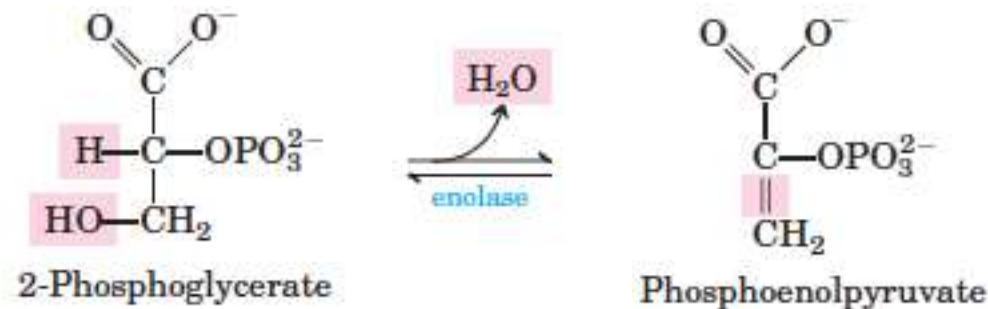


*Substrate-level phosphorylation

8. Conversion of 3-Phosphoglycerate to 2-Phosphoglycerate



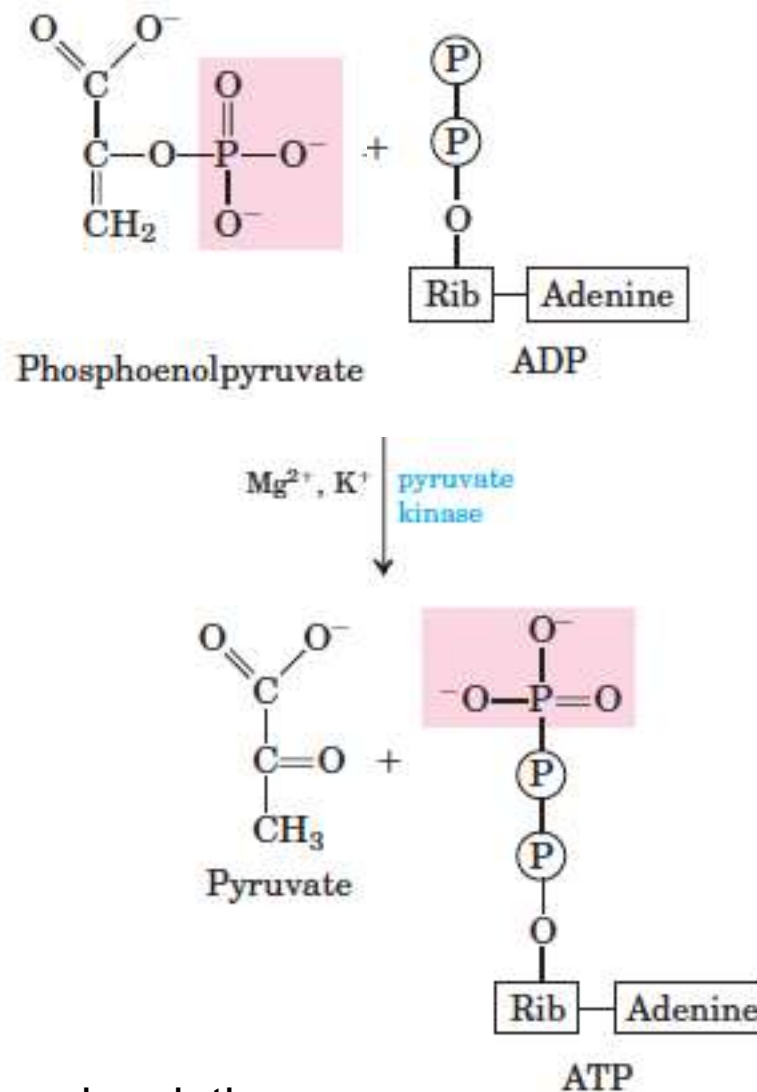
9. Dehydration of 2-Phosphoglycerate to Phosphoenolpyruvate



CLINICAL CORRELATES

The enzyme **enolase** is inhibited by **fluoride**. To prevent ongoing glycolysis in a patient's blood samples collected for sensitive **glucose tolerance tests**, blood is collected in tubes containing fluoride.

10. Transfer of the Phosphoryl Group from Phosphoenolpyruvate to ADP



*Substrate-level phosphorylation

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The product pyruvate first appears in its enol form, then tautomerizes rapidly and nonenzymatically to its keto form, which predominates at pH 7.



Summary of Glycolysis Reactions

Step	Reaction	Enzyme	Reaction type
1	Glucose + ATP $\longrightarrow$ glucose 6-phosphate + ADP + H ⁺	Hexokinase	Phosphoryl transfer
2	Glucose 6-phosphate $\rightleftharpoons$ fructose 6-phosphate	Phosphoglucose isomerase	Isomerization
3	Fructose 6-phosphate + ATP $\longrightarrow$ fructose 1,6-bisphosphate + ADP + H ⁺	Phosphofructokinase	Phosphoryl transfer
4	Fructose 1,6-bisphosphate $\rightleftharpoons$ dihydroxyacetone phosphate + glyceraldehyde 3-phosphate	Aldolase	Aldol cleavage
5	Dihydroxyacetone phosphate $\rightleftharpoons$ glyceraldehyde 3-phosphate	Triose Phosphate isomerase	Isomerization
6	Glyceraldehyde 3-phosphate + P _i + NAD ⁺ $\rightleftharpoons$ 1,3-bisphosphoglycerate + NADH + H ⁺	Glyceraldehyde 3-phosphate dehydrogenase	Phosphorylation coupled to oxidation
7	1,3-Bisphosphoglycerate + ADP $\rightleftharpoons$ 3-phosphoglycerate + ATP	Phosphoglycerate kinase	Phosphoryl transfer
8	3-Phosphoglycerate $\rightleftharpoons$ 2-phosphoglycerate	Phosphoglycerate mutase	Phosphoryl shift
9	2-Phosphoglycerate $\rightleftharpoons$ Phosphoenolpyruvate + H ₂ O	Enolase	Dehydration
10	Phosphoenolpyruvate + ADP + H ⁺ $\longrightarrow$ pyruvate + ATP	Pyruvate kinase	Phosphoryl transfer

Generation of high-energy phosphate in the Glycolytic pathway:

Pathway	Reaction Catalyzed by	Method of ~P Production	Number of ~P Formed per Mole of Glucose
Glycolysis	Glyceraldehyde-3-phosphate dehydrogenase	Respiratory chain oxidation of 2 NADH	6*
	Phosphoglycerate kinase	Phosphorylation at substrate level	2
	Pyruvate kinase	Phosphorylation at substrate level	2
			<u>10</u>
Allow for consumption of ATP by reactions catalyzed by hexokinase and phosphofructokinase			-2
			Net 8

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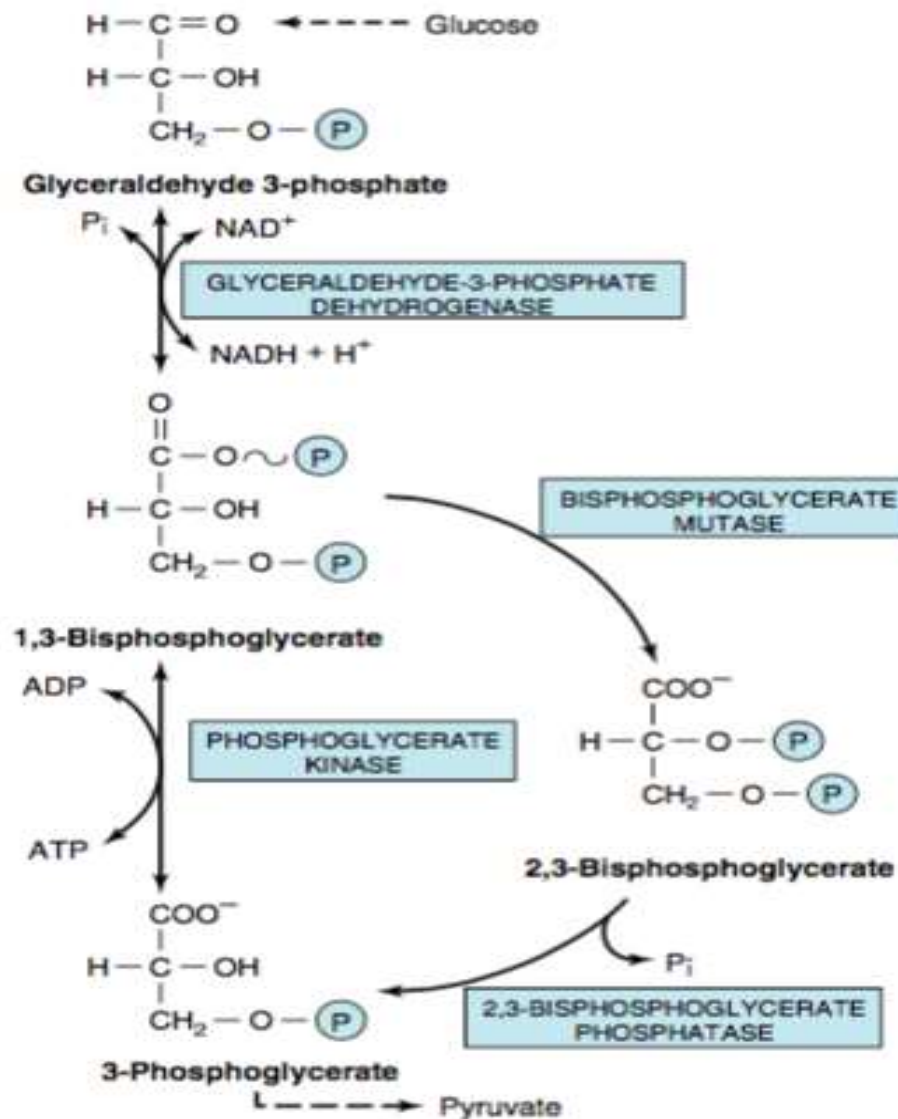
TABLE 3–8 Characteristics of Glycolysis

Entering substrates	Glucose and other monosaccharides
Enzyme location	Cytosol
Net ATP production	2 ATP formed directly per molecule of glucose entering pathway can be produced in the absence of oxygen (anaerobically)
Coenzyme production	2 NADH + 2 H ⁺ formed under aerobic conditions
Final products	Pyruvate—under aerobic conditions Lactate—under anaerobic conditions
Net reaction	
Aerobic:	Glucose + 2 ADP + 2 P _i + 2 NAD ⁺ → 2 pyruvate + 2 ATP + 2 NADH + 2 H ⁺ + 2 H ₂ O
Anaerobic:	Glucose + 2 ADP + 2 P _i → 2 lactate + 2 ATP + 2 H ₂ O

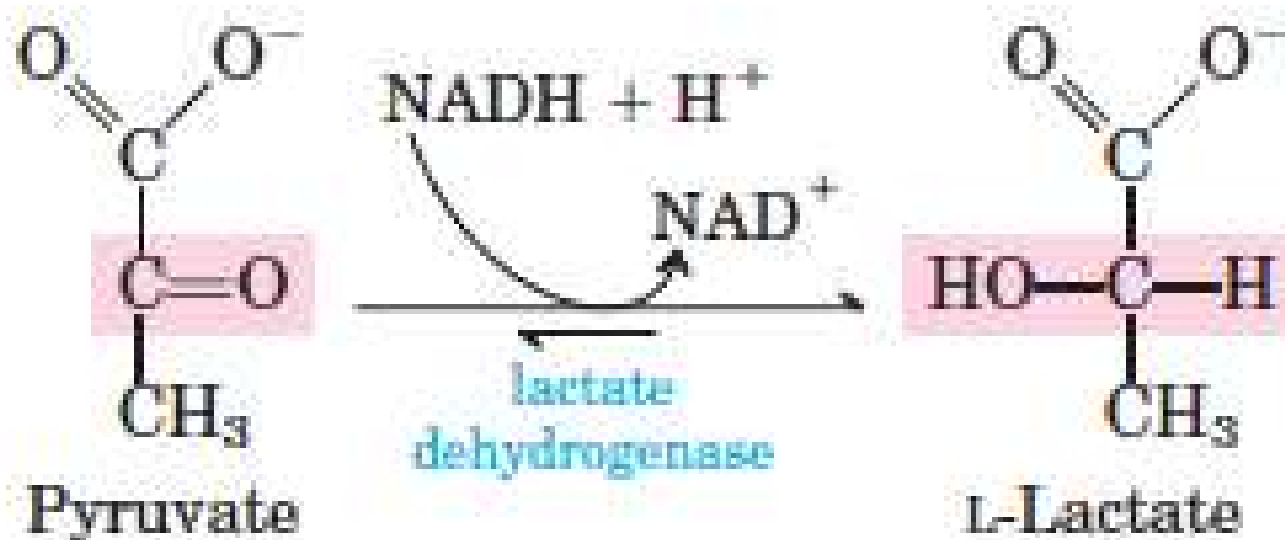
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SPECIAL REACTIONS IN RED BLOOD CELLS:

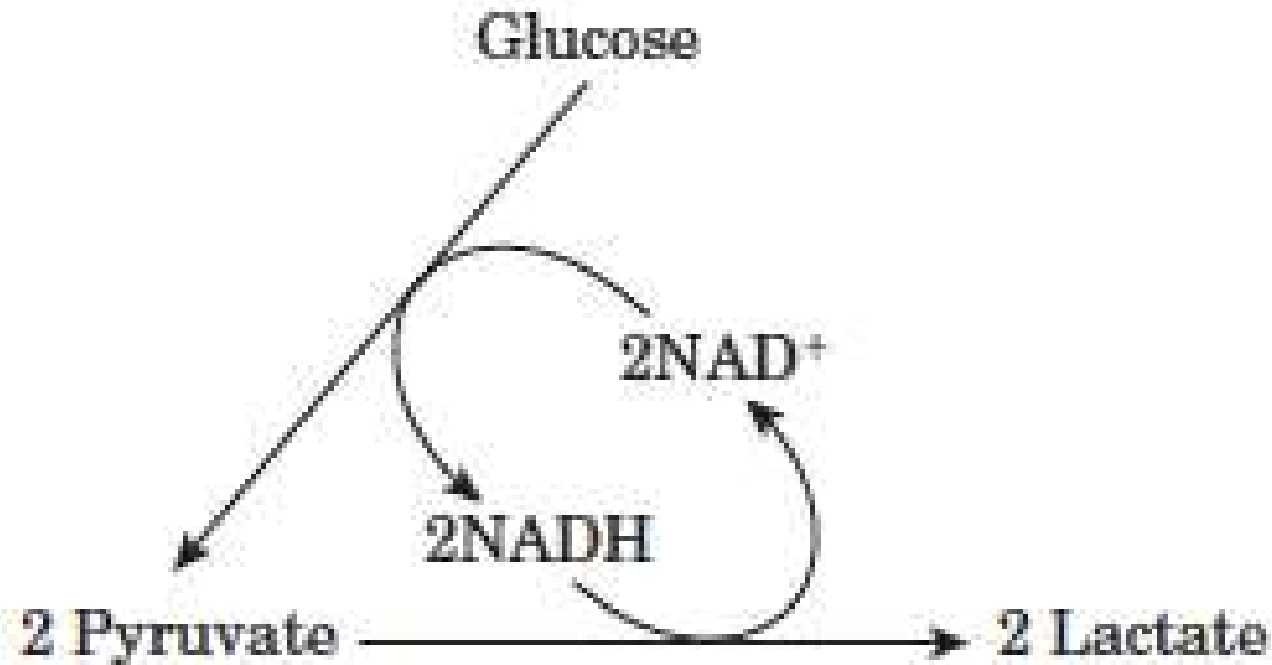
2,3-Bisphosphoglycerate pathway in Erythrocytes [Rapoport-Luebering cycle or shunt (RLC or RLS)]



Fates of Pyruvate under Anaerobic Conditions:

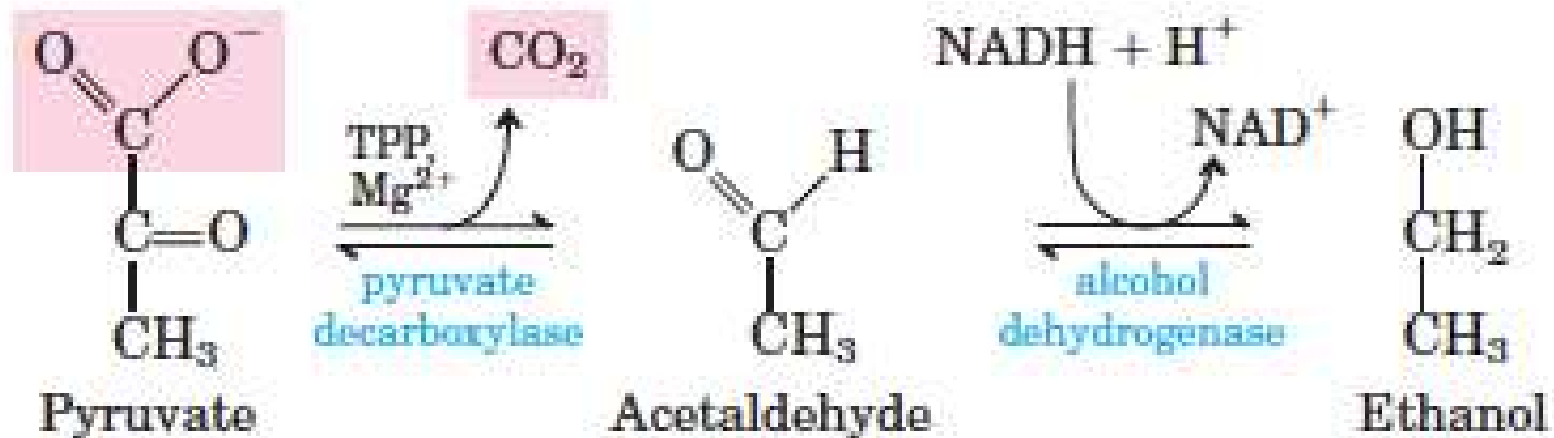


Pyruvate Is the Terminal Electron Acceptor in Lactic Acid Fermentation

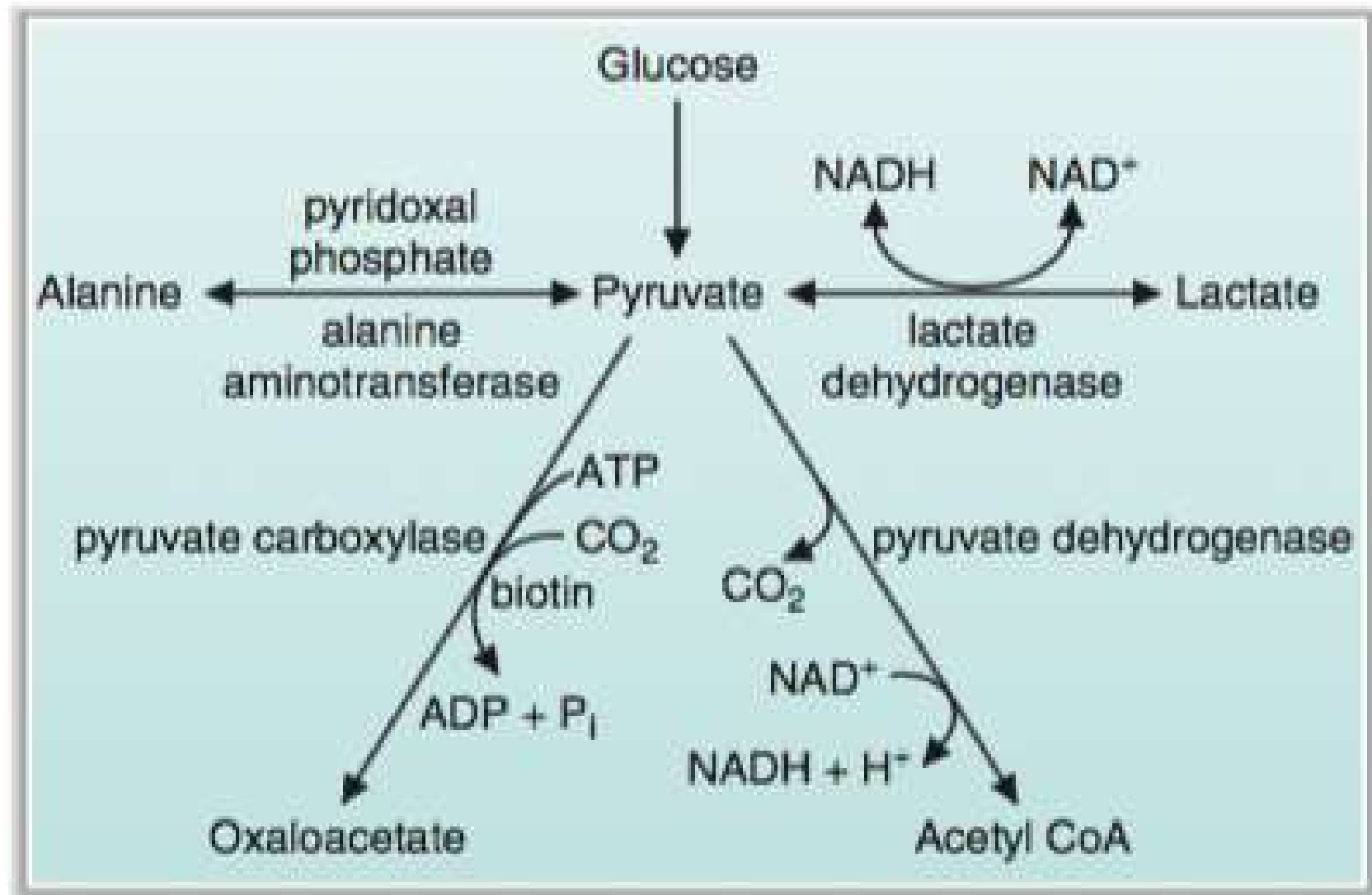


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Yeast and other microorganisms ferment glucose to ethanol and CO₂, rather than to lactate



THE FATE OF PYRUVATE



Inhibitors of Glycolysis

- Iodoacetate inhibit Glyceraldehyde-3-Phosphate Dehydrogenase
- Fluoride inhibit Enolase
 - Application: For Estimation of Blood Glucose: Sodium Fluoride: Potassium Oxalate Mixture is used
- Arsenite is toxic to 1,3-Bisphosphoglycerate Kinase step.

Regulation of Glycolysis

The rate glycolysis is regulated to meet two major cellular needs:

- (1) the production of ATP, and
- (2) the provision of building blocks for synthetic reactions.

There are three control sites in glycolysis - the reactions catalyzed by

- **hexokinase,**
- **phosphofructokinase 1, and**
- **pyruvate kinase**

These reactions are **irreversible**.

Their activities are regulated

- by the reversible binding of **allosteric effectors**
- by **covalent modification**
- by the regulation of **transcription** (change of the enzymes amounts).

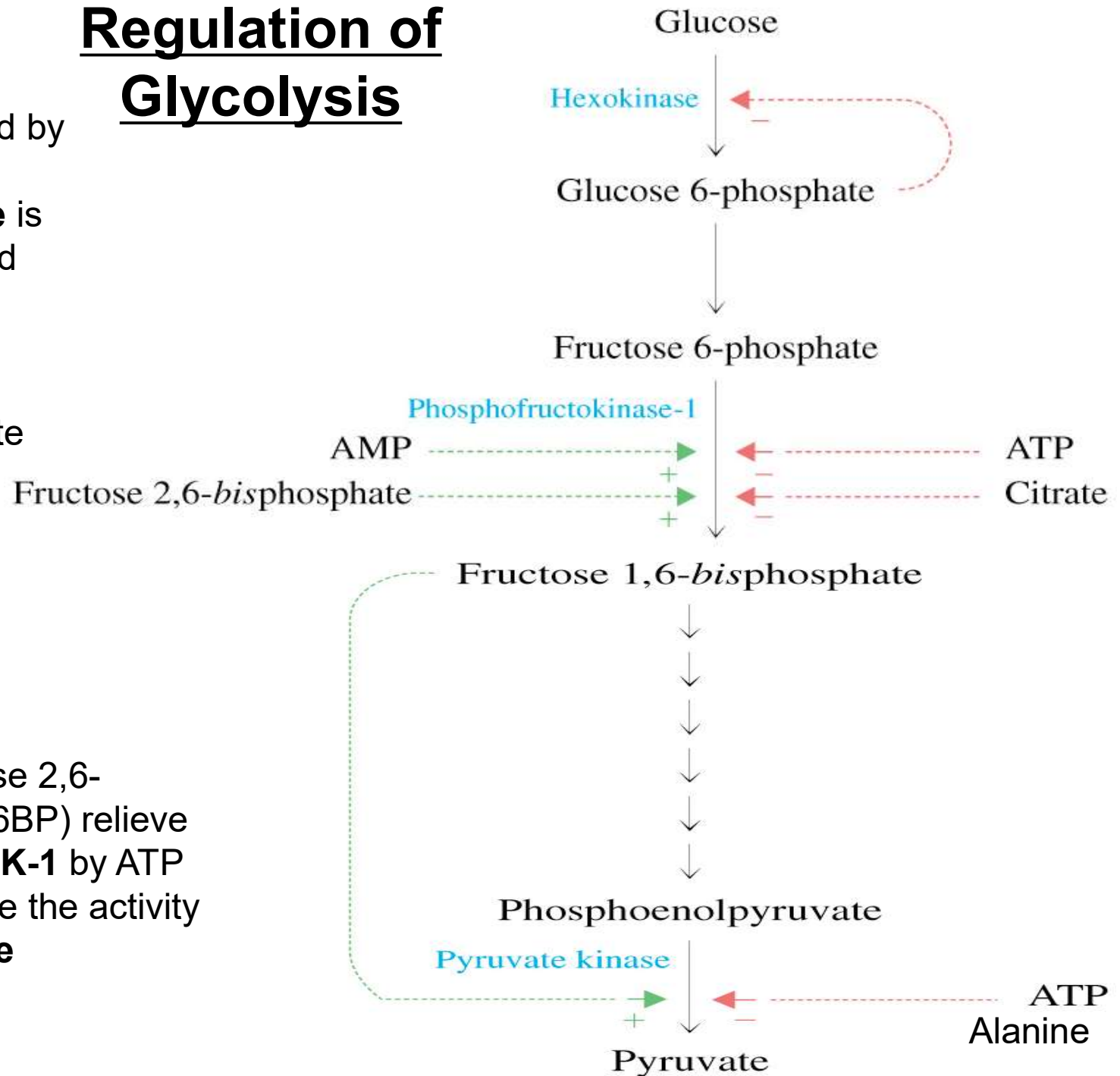
Regulation of Glycolysis

Inhibition

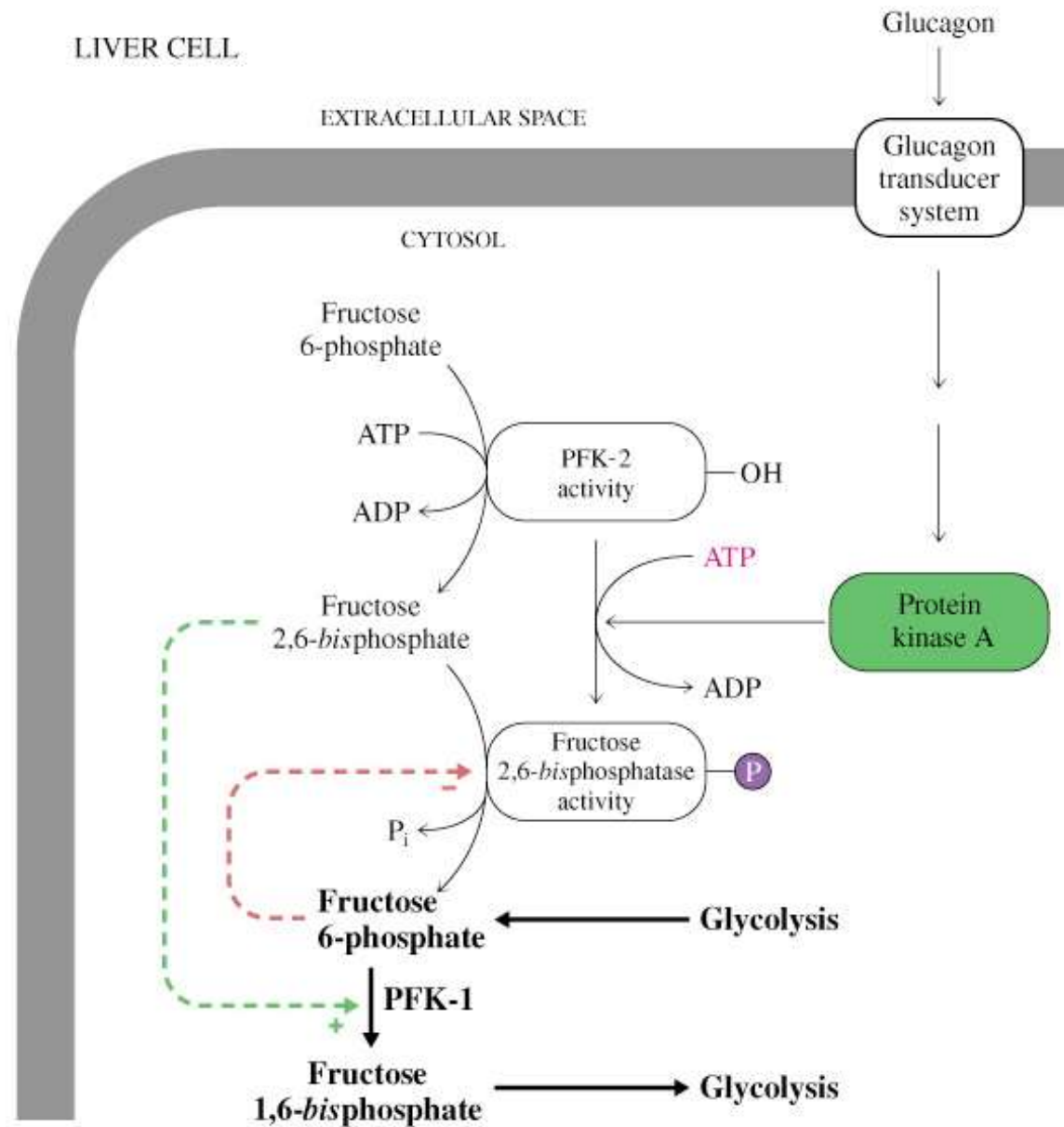
- 1) **PFK-1** is inhibited by ATP and citrate
- 2) **Pyruvate kinase** is inhibited by ATP and alanine
- 3) **Hexokinase** is inhibited by excess glucose 6-phosphate

Stimulation

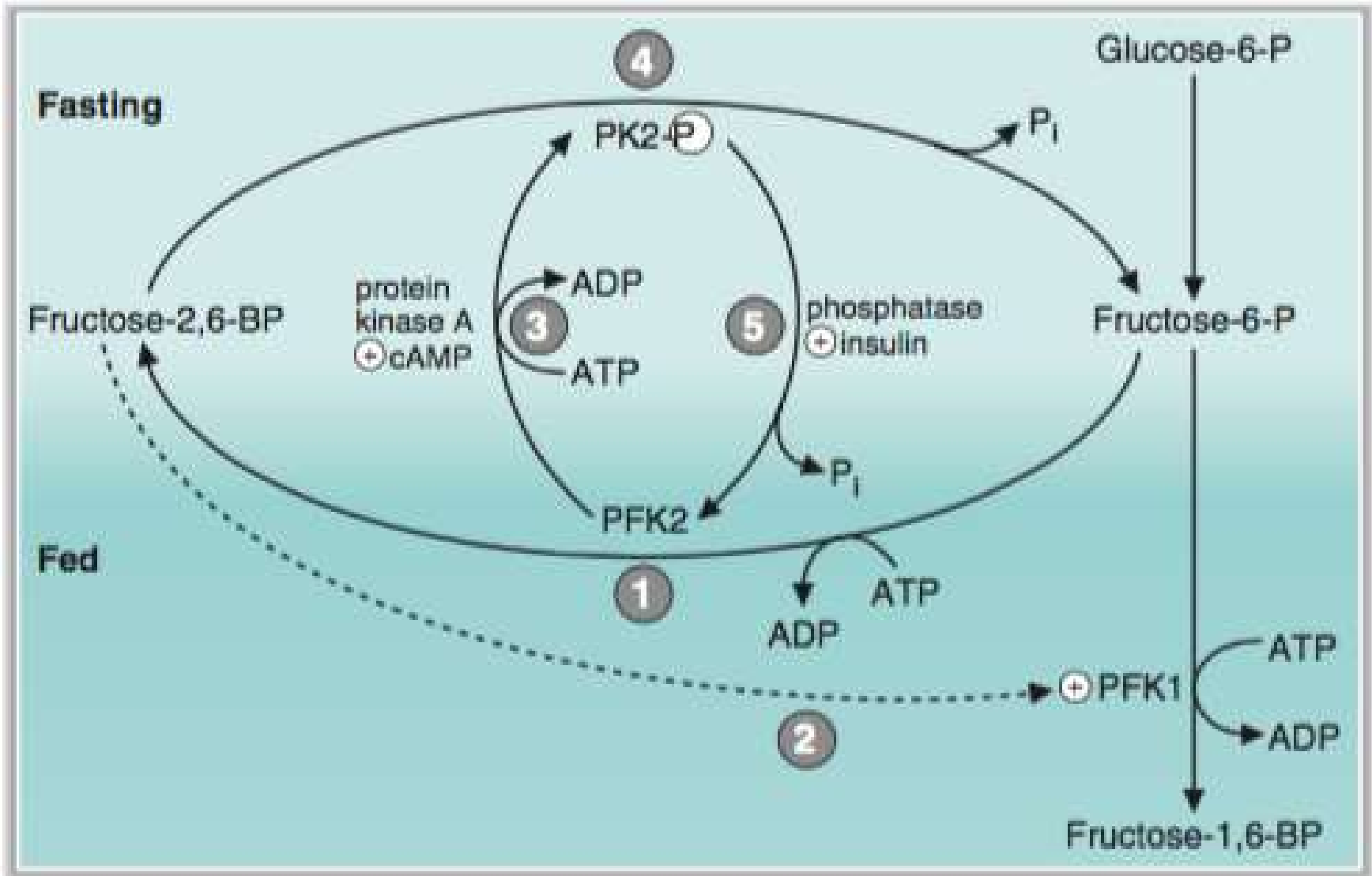
- 1) AMP and fructose 2,6-bisphosphate (F2,6BP) relieve the inhibition of **PFK-1** by ATP
- 2) F1,6BP stimulate the activity of **pyruvate kinase**



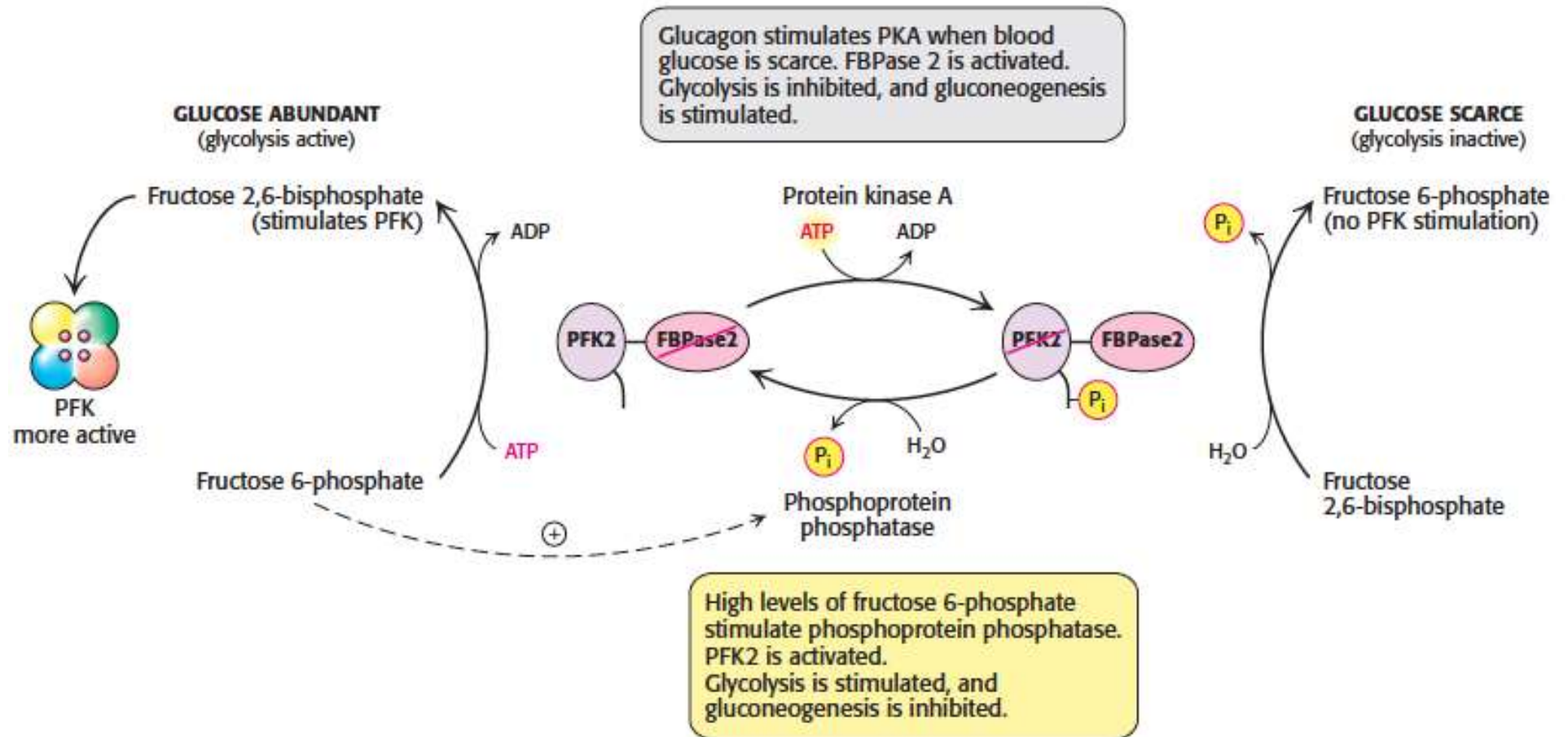
Regulation of Glycolysis by Fructose 2,6-bisphosphate



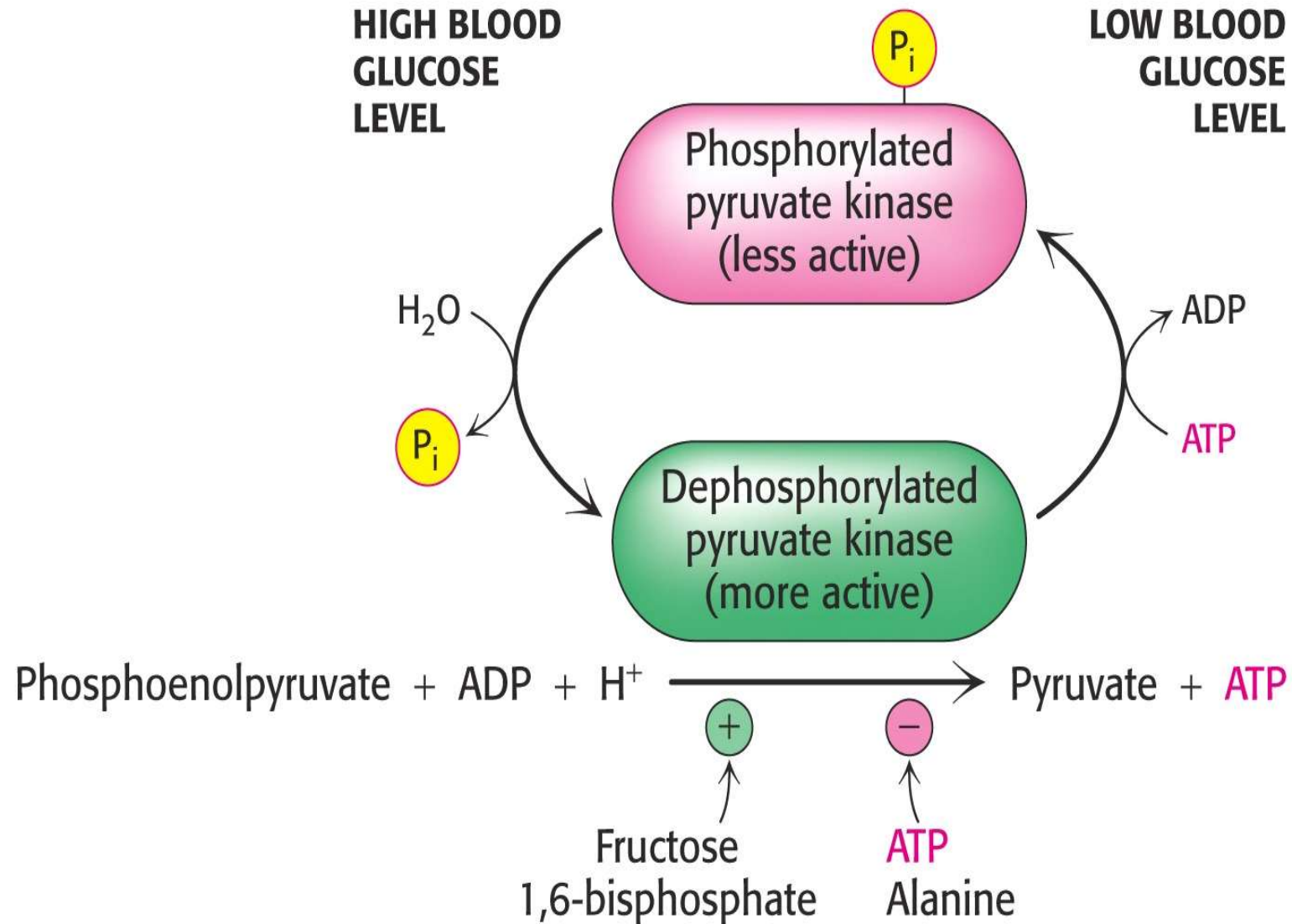
Regulation of Glycolysis by Fructose 2,6-bisphosphate



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Regulation of Pyruvate Kinase (PK)



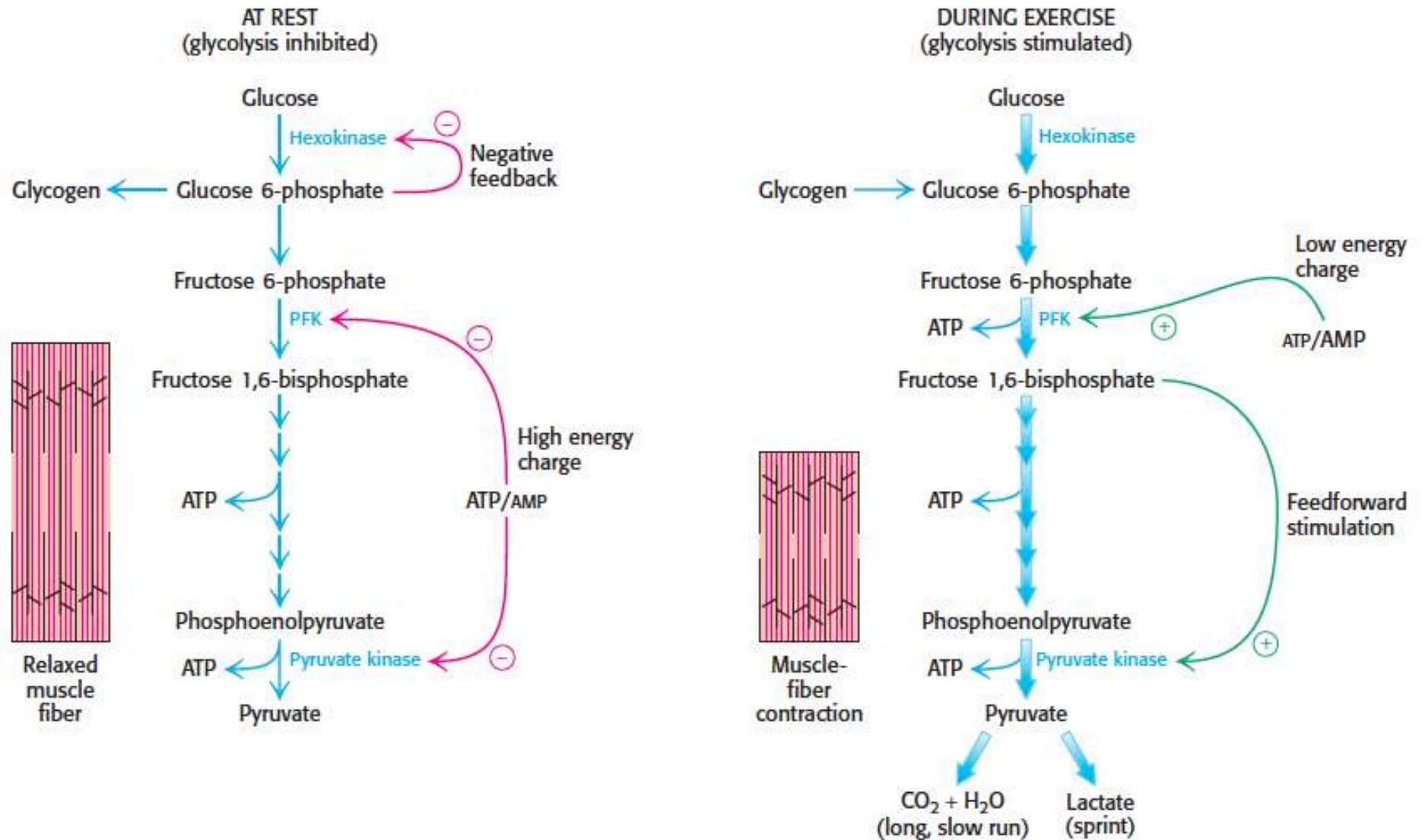
Pyruvate Kinase Deficiency

- Second most common genetic deficiency (PKLR gene; chromosome 1q22; autosomal recessive) that causes a hemolytic anemia (G6PDH deficiency is the most common).
- Characteristics include:
 - Chronic hemolysis
 - Increased 2,3-BPG and therefore a lower-than-normal oxygen affinity of HbA
 - the decrease in ATP causes the erythrocyte to lose its characteristic biconcave shape and signals its destruction in the spleen
 - decreased ion pumping by Na^+/K^+ -ATPase results in loss of ion balance and causes osmotic fragility, leading to swelling and lysis.
 - Absence of Heinz bodies (Heinz bodies are more characteristic of G6PDH deficiency)

A case file: Pyruvate Kinase Deficiency

- A 30-year-old male patient presents with a history of chronic fatigue, jaundice, and anemia. Laboratory tests reveal low hemoglobin levels, elevated reticulocyte count, and increased lactate dehydrogenase (LDH) levels.
- Biochemical changes:
 - Impaired erythrocyte glycolysis,
 - Decreased ATP production
 - Loss of membrane plasticity.
 - Premature destruction of red blood cells in the spleen or liver,
 - An accumulation of 2,3-diphosphoglycerate (2,3-DPG), causing a rightward shift in the hemoglobin-oxygen dissociation curve.
- Regular monitoring of symptoms and complications is essential to optimize the patient's quality of life and prevent further hemolytic episodes.

Regulation of glycolysis in muscle.



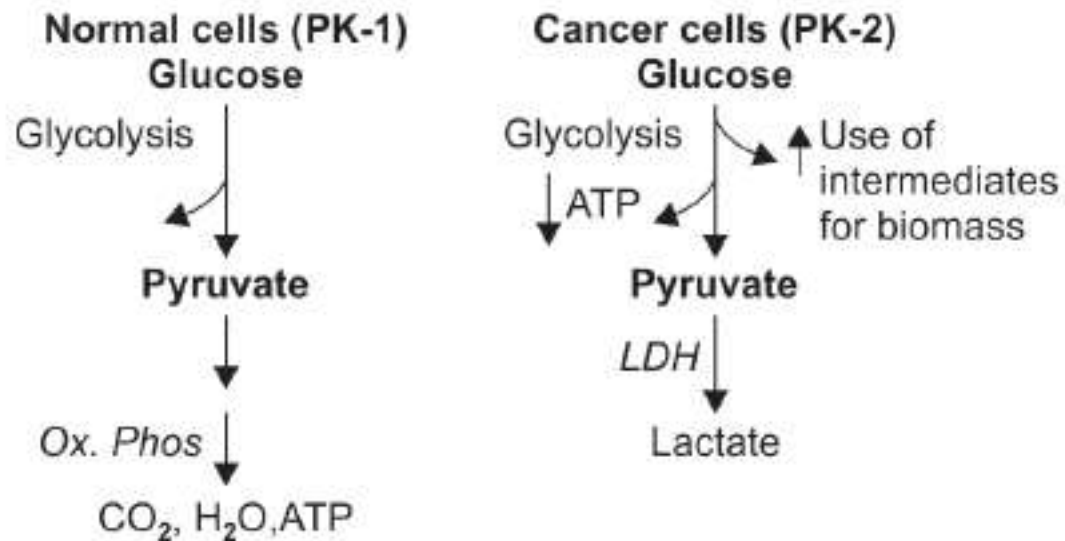
Dr Balakrishnan S, PhD.,

Cancer Cells and Warburg effect

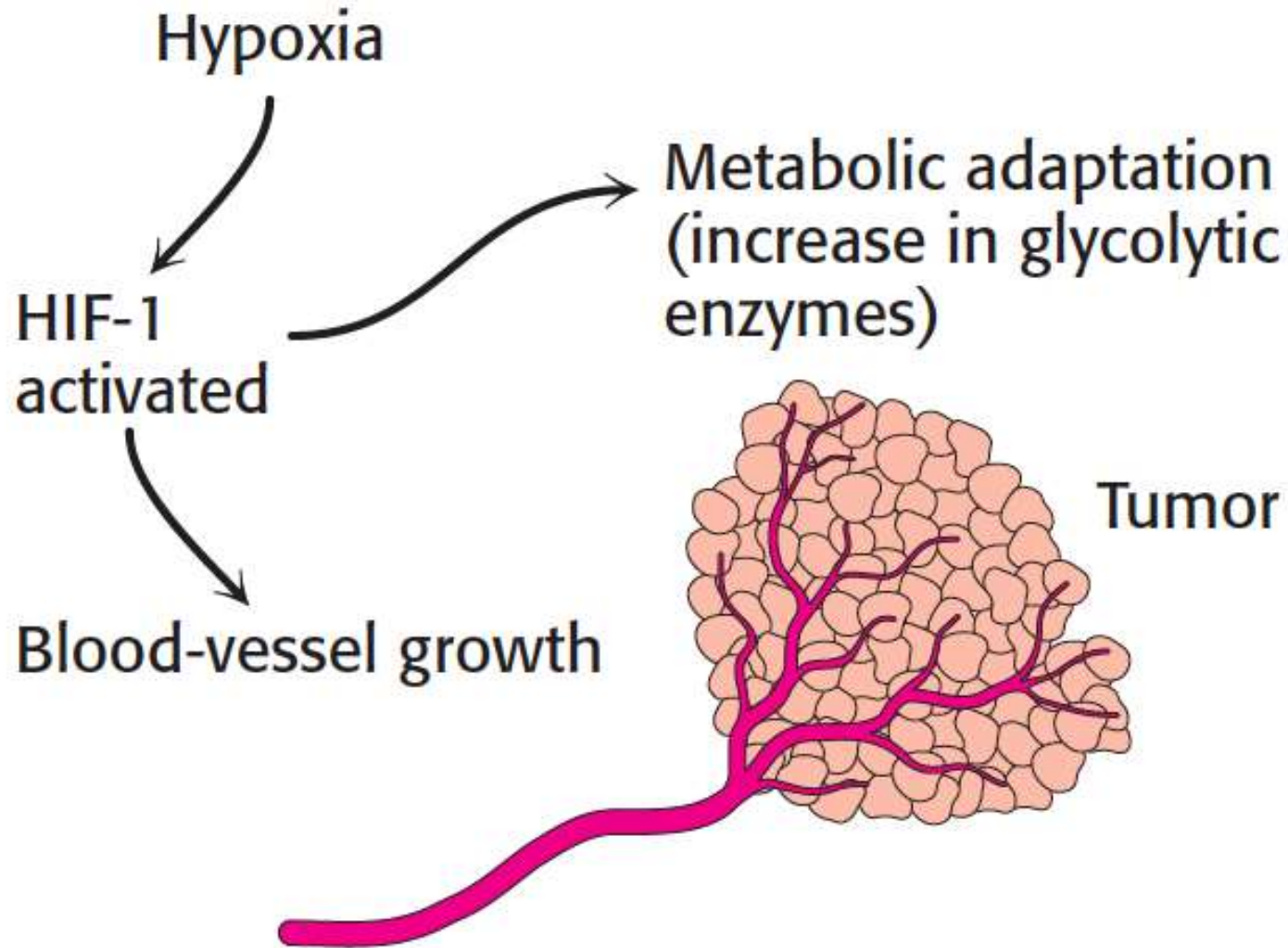
- Observed by Otto Warburg and his colleagues in 1924.
- Cancer cells take up large amount of glucose and metabolize it to lactic acid even in presence of oxygen - Warburg effect.
- Warburg hypothesis states that;
 - Increased ratio of glycolysis when compared to aerobic respiration was likely due to defect in mitochondrial respiratory chain
 - Enhanced glycolysis help cancer cells to preferentially proliferate in reduced oxygen tension
 - He also argued that switch from aerobic to anaerobic glucose metabolism was the driver of tumorigenesis.

Cancer Cells and Metabolic Reprogramming

- Rather than overt defect in mitochondrial respiratory Chain, Genetic changes in specific metabolic enzyme encoding gene
- Metabolic reprogramming of Glycolysis result in;
 - Less shuttling of glucose derived chemical energy by Glycolysis into production of ATP
 - Shunting of glucose derived chemical energy for building up of cellular biomass of proteins, lipids, etc.



Alteration of gene expression in tumors owing to hypoxia



Read more...



THANK YOU...